



control. This allows both designs.

Melexis

<https://www.melexis.com>

Ultra-low power always-on sensor IC

In common IoT designs, the microcontroller (MCU) wakes up to poll sensors, using energy even when no event is detected. The nPZero PSIC changes this by turning off the host MCU and



handling configuration and data readout from up to four sensors. The host MCU wakes up only when set rules or

thresholds are met, with tests showing up to 90% reduction in system energy use. In use cases such as smart agriculture, industrial monitoring, and asset tracking, nPZero helps devices run longer with less maintenance. The nPZero Configurator provides a GUI for setting parameters without coding, and automatic API generation reduces setup effort and software development. As energy efficiency becomes a factor in IoT, nPZero offers a hardware approach to balance battery life and sensing rate.

Nanopower

<https://nanopowersemi.com>

World's first AI-based full ISP

Chips&Media, Inc. and Visionary.ai have announced the launch of what they describe as the world's first AI-based full image signal processor (ISP). The system uses a software-defined imaging pipeline designed to improve



video quality in real time. The AI ISP replaces traditional fixed-function hardware pipelines with neural network-based processing. The companies say this represents a shift in how

imaging systems are designed. The AI ISP is designed to improve video quality across lighting and motion conditions in real time, reducing noise and blur while improving sharpness and colour accuracy. The companies also report improved computer vision performance, including more than 75% higher detection rates in low-light conditions and up to 91% reduction in false positives.

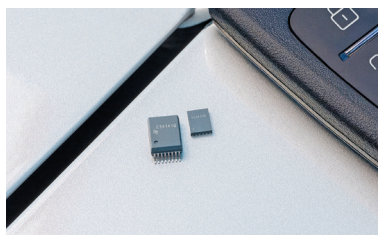
Visionary AI and Chips&Media, Inc.

<https://visionary.ai/> and

<https://chipsnmedia.com/en/index.php>

Power modules for space-saving designs

Texas Instruments (TI) has introduced isolated power modules designed to improve power density, efficiency, and



safety across applications, including data centres and electric vehicles (EVs). The UCC34141-Q1 and UCC33420 modules use TI's IsoShield technology, a multichip packaging approach that delivers up to three times the power density of discrete isolated power designs. IsoShield technology increases power density in smaller form factors, reducing system area, cost, and weight. The design can reduce solution size by up to 70% while delivering up to 2W of power, supporting compact designs for automotive, industrial, and data centre systems requiring reinforced isolation.

Texas Instruments (TI)

<https://www.ti.com>

Indoor tracking system for factories

NXP Semiconductors has announced the omlox Starter Kit, developed with SynchronicIT and Flowcate. The kit is an end-to-end hardware and software solution to evaluate and deploy real-time location systems (RTLS) based on the omlox standard. It helps customers deploy a standards-based,

interoperable RTLS system faster. It is powered by NXP's Trimension SR048 ultra-wideband (UWB) SoC, which



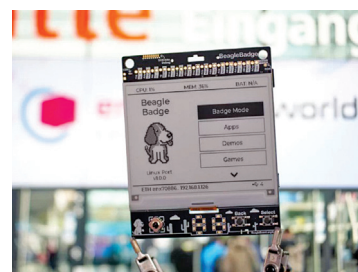
provides positioning for industrial environments, and the NXP MCX W72 wireless MCU. The kit is designed to support evaluation, prototyping, and deployment of omlox-based RTLS.

NXP Semiconductor

<https://www.nxp.com/c>

Open wearable platform for building and testing

BeagleBoard.org has announced the BeagleBoard.org BeagleBone Black (BB-Black) is an open wearable computing platform for prototyping and experimentation, combining sensing with expansion options for health, environmental monitoring, and human-machine interfaces. It includes a 4.2-inch (107mm) ePaper display with Linux drivers and support for MicroPython and LVGL, along with



Wi-Fi 6 and Bluetooth Low Energy 5.4, LoRa sub-GHz connectivity, an accelerometer and gyroscope, temperature and humidity sensing, ambient light sensing, and controls such as a 4-way joystick, buttons, a piezo buzzer, RGB LED, and dual 7-segment displays. The platform supports Linux and Zephyr with a mainline development approach and provides example applications and integrations such as Meshtastic and ActivityPub for messaging use cases, targeting learning, prototyping, and wearable and sensor-based system development.

BeagleBoard.org

<https://www.beagleboard.org>